G11Adv	Mathematics	الرياضيات	T1 · Practice

Multiple Choice Practice	تدريب اختيار من متعدد
Choose the best answer for each question.	اختر الإجابة الأنسب لكل سؤال.

Section 1 · Graph exponential growth functions

1. For $f(x) = 3^x$, which statement is true?

A) y-intercept $(1, 0)$; asymptote $y = 1$

B) y-intercept $(0, 1)$; asymptote $y = 0$

C) y-intercept $(2, 0)$; asymptote $y = -1$

D) y-intercept $(0, 2)$; asymptote $y = 2$

2. For $f(x) = 3^x$, what happens to $f(x)$ as $x \to -\infty$ and as $x \to \infty$?

A) $x \to -\infty,\ f(x) \to \infty;\ x \to \infty,\ f(x) \to 0$

B) $x \to -\infty,\ f(x) \to 0;\ x \to \infty,\ f(x) \to \infty$

C) $x \to -\infty,\ f(x) \to 0;\ x \to \infty,\ f(x) \to -\infty$

D) $x \to -\infty,\ f(x) \to \infty;\ x \to \infty,\ f(x) \to \infty$

3. Which description of the transformation matches $f(x) = -2(4)^x$?

A) $a = -2:$ compress vertically and reflect on the x-axis

B) $a = -3:$ stretch horizontally and reflect on the y-axis

C) $a = -2:$ stretch vertically and reflect on the x-axis

D) $a = -2:$ stretch vertically

Section 2 · Solve exponential inequalities

4. Bacteria A: $y = 20e^{0.079t}$. Bacteria B: $y = 52e^{0.064t}$. After how many minutes are there more A cells than B cells?

A) $t > 63.7$

B) $t > 42.3$

C) $t > 61.41$

D) $t > 49.5$

Multiple Choice Practice	تدريب اختيار من متعدد
Choose the best answer for each question.	اختر الإجابة الأنسب لكل سؤال.

Section 3 · Write logarithmic expressions in exponential form and write exponential expressions in…

5. **Write the equation in exponential form.** $\log_{15} 225 = 2$

A) $15^{225} = 2$

B) $225^2 = 15$

C) $2^{15} = 225$

D) $15^2 = 225$

6. **Write the equation in exponential form.** $\log_5 \frac{1}{25} = -2$

A) $5^{-2} = \frac{1}{25}$

B) $\left(\frac{1}{25}\right)^{-2} = 5$

C) $-5^2 = \frac{1}{25}$

D) $5^{\frac{1}{25}} = -2$

7. **Write the equation in exponential form.** $\log_3 \frac{1}{27} = -3$

A) $3^{-3} = \frac{1}{27}$

B) $\left(\frac{1}{27}\right)^{-3} = 3$

C) $-3^3 = \frac{1}{27}$

D) $3^{\frac{1}{27}} = -3$

Multiple Choice Practice	تدريب اختيار من متعدد
Choose the best answer for each question.	اختر الإجابة الأنسب لكل سؤال.

Section 3 · Write logarithmic expressions in exponential form and write exponential expressions in…

8. **Write the equation in exponential form.** $\log_4 64 = 3$

A) $64^3 = 4$ **B)** $4^3 = 64$

C) $3^4 = 64$ **D)** $64^4 = 3$

Section 4 · Solve logarithmic equations using the properties of logarithms

9. **Solve the equation.** $\log_4(2x^2 - 4) = \log_4(2x)$

A) $x = 2$ **B)** $x = -1$

C) $x = 2$ or $x = -1$ **D)** no solution

10. **Solve the equation.** $\log_5(x^2 - 6) = \log_5(x)$

A) $x = 3$ **B)** $x = -2$

C) $x = 3$ or $x = -2$ **D)** no solution

11. **Solve the equation.** $\log_3(x^2 - 8) = \log_3(2x)$

A) $x = 4$ **B)** $x = -2$

C) $x = 4$ or $x = -2$ **D)** no solution

Section 5 · Evaluate logarithmic expressions using the Change of Base Formula

12. **Express in terms of common logarithms, then approximate to the nearest ten-thousandth.** $\log_2 11$

A) 3.4594 **B)** 2.2297

C) 1.6356 **D)** 2.1322

Multiple Choice Practice	تدريب اختيار من متعدد
Choose the best answer for each question.	اختر الإجابة الأنسب لكل سؤال.

Section 5 · Evaluate logarithmic expressions using the Change of Base Formula

13. **Express in terms of common logarithms, then approximate to the nearest ten-thousandth.**
$\log_8 30$

A) 3.4594 **B)** 2.2297
C) 1.6356 **D)** 2.1322

14. **Express in terms of common logarithms, then approximate to the nearest ten-thousandth.**
$\log_5 0.4$

A) 5.6439 **B)** -0.5693
C) 0.6309 **D)** 1.3023

Section 6 · Evaluate expressions involving the natural base and natural logarithm

15. **Write the expression as a single logarithm.** $\frac{1}{3}\ln 8 - \ln 3 + \ln 9$

A) $\ln 6$ **B)** $\ln 3$
C) $2\ln 16$ **D)** $\ln 14$

16. **Write the expression as a single logarithm.** $\ln 7 - \frac{1}{3}\ln 27 + \ln 6$

A) $\ln 6$ **B)** $\ln 3$
C) $2\ln 16$ **D)** $\ln 14$

17. **Write the expression as a single logarithm.** $3\ln 3 - \ln 9$

A) $\ln 6$ **B)** $\ln 3$
C) $2\ln 16$ **D)** $\ln 14$

Multiple Choice Practice	تدريب اختيار من متعدد
Choose the best answer for each question.	اختر الإجابة الأنسب لكل سؤال.

Section 6 · Evaluate expressions involving the natural base and natural logarithm

18. **Write the expression as a single logarithm.** $4\ln 16 - \ln 256$

A) $\ln 6$ **B)** $\ln 3$

C) $2\ln 16$ **D)** $\ln 14$

19. **Write the expression as a single logarithm.** $2\ln x + 2\ln 4$

A) $\ln 8x$ **B)** $\ln 3x^2$

C) $\ln(16x^2)$ **D)** $\ln 14$

Section 7 · Simplify rational expressions

20. **Simplify the expression and state when the original expression is undefined.**

$$\frac{x(x-3)(x+6)}{x^2+x-12}$$

A) $\dfrac{x(x+6)}{x+4};\ x=-4,\ 3$

B) $\dfrac{x(x+6)}{x-4};\ x=-6,\ -4$

C) $\dfrac{x(x+6)}{x-4};\ x=-6,\ 4$

D) $\dfrac{x+6}{x+4};\ x=-4,\ 3$

Multiple Choice Practice	تدريب اختيار من متعدد
Choose the best answer for each question.	اختر الإجابة الأنسب لكل سؤال.

Section 7 · Simplify rational expressions

21. **Simplify the expression and state when the original expression is undefined.**

$$\frac{y^2(y^2+3y+2)}{2y(y-4)(y+2)}$$

A) $\frac{y(y-1)}{2(y-4)};\ y=-2,0,4$

B) $\frac{y(y-1)}{2(y-4)};\ y=0,2,4$

C) $\frac{y(y+1)}{2(y-4)};\ y=-2,0,4$

D) $\frac{y(y+1)}{(y+4)};\ y=-2,4$

Section 8 · Simplify complex algebraic fractions including rational expressions

22. **Simplify.** $\frac{3}{x}+\frac{5}{y}$

A) $\frac{3x+5y}{xy}$

B) $\frac{5x+3y}{xy}$

C) $\frac{5x+5x}{x}$

D) $\frac{5x+3y}{y}$

Multiple Choice Practice	تدريب اختيار من متعدد
Choose the best answer for each question.	اختر الإجابة الأنسب لكل سؤال.

Section 8 · Simplify complex algebraic fractions including rational expressions

23. Simplify. $\frac{3}{8p^2r} + \frac{5}{4p^2r}$

A) $\frac{15}{8pr}$

B) $\frac{13}{8r}$

C) $\frac{13}{8p^2r}$

D) $\frac{15}{8pr^2}$

24. Simplify. $\frac{2c-7}{3} + 4$

A) $\frac{2c+5}{3}$

B) $\frac{5c+2}{3}$

C) $\frac{2c-5}{3}$

D) $\frac{2c-2}{3}$

Multiple Choice Practice	تدريب اختيار من متعدد
Choose the best answer for each question.	اختر الإجابة الأنسب لكل سؤال.

Section 9 · Graph transformations of reciprocal functions

25. Find the vertical and oblique asymptotes of $f(x) = \dfrac{(x-4)^2}{x+2}$.

A) $x = 5;\ f(x) = x + 11$

B) $x = -2;\ f(x) = 6x - 8$

C) $x = -2;\ f(x) = x - 10$

D) $x = 2;\ f(x) = 2x + 11$

26. Find the zeros and asymptotes of $f(x) = \dfrac{6x^2 + 4x + 2}{x+2}$. Which pair is correct?

A) no zeros; $x = -2;\ f(x) = 6x - 8$

B) $x = 0$ or $x = -3.5;\ x = -2;\ f(x) = 6x - 8$

C) $x = -3;\ x = 2;\ f(x) = 2x + 11$

D) $x = 4;\ x = 5;\ f(x) = x + 11$

Section 10 · Solve rational equations

27. Solve the equation. $\dfrac{2x+3}{x+1} = \dfrac{3}{2}$

A) $x = -3$

B) $x = 9$

C) $x = 7$

D) $x = -5$

Multiple Choice Practice	تدريب اختيار من متعدد
Choose the best answer for each question.	اختر الإجابة الأنسب لكل سؤال.

Section 10 · Solve rational equations

28. **Solve the equation.** $\frac{-12}{y} = y - 7$

A) $y = 3$ and 4 **B)** $y = 2$

C) $y = -8$ and 3 **D)** $y = 4$

29. **Solve the equation.** $\frac{9}{x-7} - \frac{7}{x-6} = \frac{13}{x^2 - 13x + 42}$

A) $x = -3$ **B)** $x = 9$

C) $x = 7$ **D)** $x = -5$

Section 11 · Classify and determine study types

30. **Identify the question or sample as biased or unbiased: "Every tenth student at band camp is asked to name his or her favorite band for a survey about the campers."**

A) Biased **B)** Unbiased

31. **Identify the question or sample as biased or unbiased: "Which is your favorite type of music, pop or country?"**

A) Biased **B)** Unbiased

Multiple Choice Practice	تدريب اختيار من متعدد
Choose the best answer for each question.	اختر الإجابة الأنسب لكل سؤال.

Section 11 · Classify and determine study types

32. **Identify the question or sample as biased or unbiased: "Don't you agree that employees should pack their lunch?"**

A) Biased **B)** Unbiased

33. **Identify the question or sample as biased or unbiased: "Are you a member of any after-school clubs?"**

A) Biased **B)** Unbiased

Section 12 · Find and compare experimental and theoretical probabilities

34. **A spinner with 4 equal sections is spun 100 times: Red 35, Blue 38, Green 13, Yellow 14. Find the THEORETICAL probability of blue.**

A) 25% **B)** 50%
C) 58.4% **D)** 38%

35. **Same spinner (Red 35, Blue 38, Green 13, Yellow 14 in 100 spins). Find the EXPERIMENTAL probability of blue.**

A) 25% **B)** 50%
C) 58.4% **D)** 38%

36. **A coin is flipped 125 times: 73 heads, 52 tails. Find the THEORETICAL probability of heads.**

A) 25% **B)** 50%
C) 58.4% **D)** 38%

37. **A coin is flipped 125 times: 73 heads, 52 tails. Find the EXPERIMENTAL probability of heads.**

A) 25% **B)** 50%
C) 58.4% **D)** 38%

Multiple Choice Practice	تدريب اختيار من متعدد
Choose the best answer for each question.	اختر الإجابة الأنسب لكل سؤال.

Section 12 · Find and compare experimental and theoretical probabilities

38. **A fair die is rolled 150 times and a 3 appears 27 times. Find the EXPERIMENTAL probability of rolling a 3.**

A) 27.4% **B)** 16.7%

C) 18% **D)** 29.1%

39. **A card is drawn from a standard deck and returned, 120 times: Heart 28, Diamond 37, Spade 34, Club 21. Which statement is true?**

A) Theoretical P(heart) is less than experimental P(heart)

B) Theoretical P(diamond) is less than experimental P(diamond)

C) Experimental P(club) is greater than theoretical P(club)

D) Experimental P(heart) equals experimental P(diamond)

Section 14 · Describe a data distribution by its center, spread, and overall shape

40. **Race times (s): 57.1, 55.9, 59.3, 54.9, 54.6, 50.3, 55.2, 53.5. Find the standard deviation of the data.**

A) $\sigma = 3.25$ **B)** $\sigma = 3.45$

C) $\sigma = 2.45$ **D)** $\sigma = 1.45$

41. **Prices (AED): 50, 165, 55, 79, 84, 68, 38, 42. Find the standard deviation of the data.**

A) $\sigma = 34.21$ **B)** $\sigma = 38.22$

C) $\sigma = 31.22$ **D)** $\sigma = 41.15$

Multiple Choice Practice	تدريب اختيار من متعدد
Choose the best answer for each question.	اختر الإجابة الأنسب لكل سؤال.

Section 15 · Draw and find angles in standard position

42. Rewrite the measure in the other unit. 18°

A) $-\frac{2\pi}{5}$

B) $\frac{\pi}{10}$

C) $\frac{\pi}{30}$

D) $-\frac{\pi}{2}$

43. Rewrite the measure in the other unit. -72°

A) $-\frac{2\pi}{5}$

B) $\frac{\pi}{10}$

C) $\frac{\pi}{30}$

D) $-\frac{\pi}{2}$

44. Rewrite the measure in the other unit. $\frac{5\pi}{2}$

A) -810°

B) 450°

C) 720°

D) -105°

Multiple Choice Practice	تدريب اختيار من متعدد
Choose the best answer for each question.	اختر الإجابة الأنسب لكل سؤال.

Section 15 · Draw and find angles in standard position

45. Rewrite the measure in the other unit. $-\frac{7\pi}{12}$

A) $-810°$

B) $450°$

C) $720°$

D) $-105°$

Section 16 · Find values of trigonometric ratios

46. Find the exact value. $\tan 330°$

A) $-\frac{1}{2}$

B) $-\frac{\sqrt{3}}{3}$

C) 0.715

D) 0.134

47. Find the exact value. $\cos\left(-\frac{11\pi}{4}\right)$

A) -0.156

B) $\sqrt{2}$

C) $-\frac{\sqrt{2}}{2}$

D) $\sqrt{3}$

Multiple Choice Practice	تدريب اختيار من متعدد
Choose the best answer for each question.	اختر الإجابة الأنسب لكل سؤال.

Section 16 · Find values of trigonometric ratios

48. **Find the exact value.** $\cot 30^\circ$

A) -0.156 **B)** $\sqrt{2}$

C) $-\frac{\sqrt{2}}{2}$ **D)** $\sqrt{3}$

49. **Find the exact value.** $\sin(-150^\circ)$

A) $-\frac{1}{2}$ **B)** $-\frac{\sqrt{3}}{3}$

C) 0.715 **D)** -1

Section 17 · Use the properties of periodic functions to evaluate trigonometric functions

50. **The terminal side of angle θ in standard position meets the unit circle at** $P\left(-\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$.

Find $\sin\theta$ **and** $\cos\theta$.

A) $\sin\theta = -1,\ \cos\theta = 0$

B) $\sin\theta = 0,\ \cos\theta = -1$

C) $\sin\theta = \frac{1}{2},\ \cos\theta = -\frac{\sqrt{3}}{2}$

D) $\sin\theta = \frac{\sqrt{5}}{3},\ \cos\theta = -\frac{2}{3}$

Multiple Choice Practice	تدريب اختيار من متعدد
Choose the best answer for each question.	اختر الإجابة الأنسب لكل سؤال.

Section 17 · Use the properties of periodic functions to evaluate trigonometric functions

51. The terminal side of angle θ in standard position meets the unit circle at $P(0, -1)$. Find $\sin\theta$ and $\cos\theta$.

A) $\sin\theta = -1,\ \cos\theta = 0$

B) $\sin\theta = 0,\ \cos\theta = -1$

C) $\sin\theta = \frac{1}{2},\ \cos\theta = -\frac{\sqrt{3}}{2}$

D) $\sin\theta = \frac{\sqrt{5}}{3},\ \cos\theta = -\frac{2}{3}$

52. The terminal side of angle θ in standard position meets the unit circle at $P\left(-\frac{2}{3}, \frac{\sqrt{5}}{3}\right)$.

Find $\sin\theta$ and $\cos\theta$.

A) $\sin\theta = -1,\ \cos\theta = 0$

B) $\sin\theta = 0,\ \cos\theta = -1$

C) $\sin\theta = \frac{1}{2},\ \cos\theta = -\frac{\sqrt{3}}{2}$

D) $\sin\theta = \frac{\sqrt{5}}{3},\ \cos\theta = -\frac{2}{3}$

Multiple Choice Practice	تدريب اختيار من متعدد
Choose the best answer for each question.	اختر الإجابة الأنسب لكل سؤال.

Section 18 · Use sinusoidal functions to solve problems

53. A spring oscillates by $y = 40\cos \pi t$ (y in cm, t in seconds). Find the period.

A) $\frac{1}{60}$

B) 2

C) π

D) $\frac{2\pi}{3}$

54. For $y = 40\cos \pi t$, find the range.

A) $[-25, 25]$

B) $[-100\sqrt{2}, 100\sqrt{2}]$

C) $[-40, 40]$

D) $\left[-\frac{\pi}{5}, \frac{\pi}{5}\right]$

Section 19 · Describe and graph trigonometric functions other than sine, cosine, and tangent

55. For $y = \cos\theta + 3$, what are the amplitude and the midline?

A) $a = \frac{1}{3};\ y = 3$

B) $a = 2;\ y = \frac{1}{2}$

C) $a = 1;\ y = 3$

D) $a = 1;\ y = -1$

Multiple Choice Practice	تدريب اختيار من متعدد
Choose the best answer for each question.	اختر الإجابة الأنسب لكل سؤال.

Section 19 · Describe and graph trigonometric functions other than sine, cosine, and tangent

56. For $y = \cos\theta + 3$, find the range.

A) $-\infty \le f(\theta) \le \infty$

B) $-6 \le f(\theta) \le -2$

C) $2 \le f(\theta) \le 4$

D) $-7 \le f(\theta) \le -3$

57. For $y = 2\sin\theta - 4$, which statement is correct?

A) $a = 2;\ \text{period } 360°;\ \text{midline } y = -4$

B) $a = \frac{1}{3};\ \text{period } 270°;\ \text{midline } y = 3$

C) $a = 1;\ \text{period } 360°;\ \text{midline } y = -1$

D) $a = 2;\ \text{period } 180°;\ \text{midline } y = 4$

58. For $y = 2\sin\theta - 4$, find the range.

A) $-\infty \le f(\theta) \le \infty$

B) $-6 \le f(\theta) \le -2$

C) $-7 \le f(\theta) \le -3$

D) $2 \le f(\theta) \le 4$

Section 20 · Find values of angle measures by using inverse trigonometric functions

59. Find the value in degrees. $\tan^{-1}\sqrt{3}$

A) $60°$

B) $30°$

C) $45°$

D) $-60°$

Multiple Choice Practice	تدريب اختيار من متعدد
Choose the best answer for each question.	اختر الإجابة الأنسب لكل سؤال.

Section 20 · Find values of angle measures by using inverse trigonometric functions

60. Find the value in radians. $\cos^{-1}\left(\frac{\sqrt{2}}{2}\right)$

A) $\frac{\pi}{4}$

B) $\frac{\pi}{6}$

C) $\frac{\pi}{3}$

D) $-\frac{\pi}{3}$

Section 21 · Evaluate expressions involving the natural base and natural logarithm

61. Which description matches $f(x) = -2e^{x-3} + 2$?

A) $a = -2$ (compress vertically, reflect on x-axis); $h = -3$

B) $a = -2$ (stretch vertically, reflect on x-axis); $h = 3$ (right); $k = 2$ (up)

C) $a = -2$ (stretch vertically, reflect on x-axis); $h = 2$; $k = 3$

D) $a = -2$ (compress vertically, reflect on x-axis); $k = 3$; $h = 2$

Multiple Choice Practice	تدريب اختيار من متعدد
Choose the best answer for each question.	اختر الإجابة الأنسب لكل سؤال.

Section 22 · Use logarithms to solve problems involving exponential growth and decay

62. An experiment starts with 20 bacteria A cells. After 45 minutes there are 710. Which equation models the number of cells y after t minutes? ($y = ae^{kt}$)

A) $y = 52e^{0.012t}$

B) $y = 25e^{0.079t}$

C) $y = 20e^{0.079t}$

D) $y = 20e^{0.012t}$

63. Using $y = 20e^{0.079t}$, after how many minutes will there be 1000 bacteria A cells?

A) $t \approx 63.7$

B) $t \approx 42.3$

C) $t \approx 61.41$

D) $t \approx 49.5$

64. In 2000 the world population was 6.124 billion; in 2005 it was 6.515 billion. Which equation models the population y (billions) t years after 2000?

A) $y = 2.141e^{0.0154t}$

B) $y = 6.124e^{0.0124t}$

C) $y = 6.124e^{0.0164t}$

D) $y = 8.124e^{0.0124t}$

65. Using $y = 6.124e^{0.0124t}$, in which year does the world population reach 7.5 billion?

A) 2015

B) 2016

C) 2010

D) 2019

Multiple Choice Practice	تدريب اختيار من متعدد
Choose the best answer for each question.	اختر الإجابة الأنسب لكل سؤال.

Section 23 · Graph rational functions with oblique asymptotes and point discontinuity

66. **The graph of $g(x) = \frac{a}{x-h} + k$ has vertical asymptote x = −4, horizontal asymptote y = 0, passing through (5, 1). Which function is it?**

A) $g(x) = \frac{9}{x+4}$

B) $g(x) = \frac{9}{x-4}$

C) $g(x) = \frac{5}{x+4}$

D) $g(x) = \frac{-9}{x+4}$

67. **The graph of $g(x) = \frac{a}{x-h} + k$ has vertical asymptote x = −4, horizontal asymptote y = −2, passing through (−5, 4). Which function is it?**

A) $g(x) = \frac{6}{x+4} - 2$

B) $g(x) = \frac{-6}{x+4} - 2$

C) $g(x) = \frac{-6}{x-4} + 2$

D) $g(x) = \frac{4}{x+4} - 2$

Multiple Choice Practice	تدريب اختيار من متعدد
Choose the best answer for each question.	اختر الإجابة الأنسب لكل سؤال.

Section 23 · Graph rational functions with oblique asymptotes and point discontinuity

68. The graph of $g(x) = \dfrac{a}{x-h} + k$ has vertical asymptote x = 2, horizontal asymptote y = 2, passing through (7, 1). Which function is it?

A) $g(x) = \dfrac{5}{x-2} + 2$

B) $g(x) = \dfrac{-5}{x-2} + 2$

C) $g(x) = \dfrac{-5}{x+2} + 2$

D) $g(x) = \dfrac{1}{x-2} + 2$

Section 24 · (a) Find the area under normal distribution curves

69. Find the z-value. $X = 24,\ \mu = 19,\ \sigma = 3.8$

A) 0.76 **B)** 1.32

C) 1.82 **D)** -1.32

70. Find the z-value. $X = 106.3,\ \mu = 88.8,\ \sigma = 9.6$

A) 0.76 **B)** 1.32

C) 1.82 **D)** -1.32

71. Find the z-value. $\sigma = 9.8,\ X = 55.4,\ \mu = 68.34$

A) 0.76 **B)** 1.32

C) 1.82 **D)** -1.32

Multiple Choice Practice	تدريب اختيار من متعدد
Choose the best answer for each question.	اختر الإجابة الأنسب لكل سؤال.

Section 25 · (a) Convert between degrees and radian measures and find arc lengths by using central angles

72. **Find the period of** $y = \tan 5x$.

A) 180°

B) 36°

C) 72°

D) 900°

73. **Find the asymptotes of** $y = \tan 5x$ **(n is an integer).**

A) $x = (18 + 36n)^\circ$

B) $x = (90 + 180n)^\circ$

C) $x = (36n)^\circ$

D) $x = (9 + 18n)^\circ$